

DOCUMENT 1 — PROJECT DECISION LOG & TECHNICAL RATIONALE

File name: 01_Decision_Log_Technical_Rationale

PS 26083 — Decision Log & Technical Rationale

Project Objective

Our objective is to develop a localized heat-health early warning framework that goes beyond conventional temperature-based heatwave alerts. Instead of asking only how hot the weather will be, the proposed system aims to estimate the thermal burden experienced by people over multiple consecutive days, identify areas with higher vulnerability and lower response capacity, and translate the resulting risk information into actionable recommendations for authorities.

The project is designed as a progressive pipeline. The first stage uses real historical weather data to calculate human thermal stress. The second stage introduces nighttime thermal load and cumulative heat burden. The third stage adds spatial vulnerability and response-capacity information. The final stage is designed to support health-impact and mortality-risk estimation once appropriate historical health outcome data becomes available.

Decision 1 — Pilot City: Ahmedabad

Ahmedabad has been selected as the initial pilot city for demonstrating the proposed system. The choice is based on the relevance of extreme heat to the city, the availability of an established heat-health action context, and the practical suitability of demonstrating localized interventions at an urban level. Ahmedabad allows us to connect meteorological information with ward-level population characteristics, healthcare accessibility and response planning in a realistic pilot scenario.

The Ahmedabad implementation should be treated as a pilot rather than a limitation of the system. The underlying methodology is designed so that the same pipeline can eventually be adapted to other Indian cities and regions.

Decision 2 — Historical Weather Data: ERA5

ERA5 reanalysis data has been selected for historical weather analysis because the project requires consistent, hourly meteorological information over historical periods. The historical dataset will provide variables required for deriving thermal-stress indicators, including near-surface temperature, dew point, wind and radiation-related information.

ERA5 will primarily be used for historical analysis, feature engineering, baseline construction and retrospective testing. It is important to distinguish ERA5 reanalysis from operational forecasting. ERA5 represents historical/reconstructed atmospheric conditions and is therefore not being presented as the operational 3–5 day forecast source.

Decision 3 — Operational Forecast Source

For the final early-warning pipeline, operational forecast information from appropriate Indian meteorological sources such as IMD/NCMRWF is intended to replace the historical weather input. The same feature-engineering and risk pipeline can then process forecast weather for the coming 3–5 days.

If operational forecast access is not available during the prototype stage, the system will demonstrate forecast readiness through historical replay using known weather periods. Such a demonstration will be explicitly labelled as a prototype/replay and will not be presented as a live operational forecast.

Decision 4 — Primary Thermal-Stress Metric: UTCI

UTCI has been selected as the primary human thermal-stress metric because the problem statement requires moving beyond dry-bulb temperature and considering multiple environmental conditions affecting human thermal stress. UTCI provides a standardized way of representing thermal stress using meteorological conditions including air temperature, humidity, wind and mean radiant temperature.

UTCI is therefore used as the main representation of human thermal stress in the proposed pipeline. The project does not claim to have invented UTCI; the innovation lies in how the thermal-stress information is combined with nighttime thermal load, cumulative burden, spatial vulnerability and response capacity to create an actionable early-warning framework.

Decision 5 — Secondary Metric: WBGT

WBGT will be considered as a secondary comparison and validation metric because it is directly relevant to occupational and outdoor heat-stress assessment and is explicitly mentioned in the problem statement.

UTCI remains the primary thermal metric. WBGT is not being treated as a second independent prediction system. Instead, a comparison can demonstrate that the proposed thermal-stress pipeline is consistent with another established heat-stress indicator.

Decision 6 — Nighttime Metric: NCTL

We propose a metric called **Nighttime Cumulative Thermal Load (NCTL)** to capture persistent nighttime thermal exposure. The motivation is that extreme heat can continue after daytime peak temperatures, reducing the opportunity for nighttime cooling.

NCTL is intentionally described as a thermal-load indicator rather than a medically validated physiological recovery measurement. The project therefore does not claim that NCTL directly measures human physiological recovery. Its purpose in the prototype is to quantify the duration and magnitude of nighttime thermal conditions above a locally derived baseline.

The proposed formulation is:

$$\text{NCTL}(t) = \sum \max(0, \text{UTCI}_{\text{night}}(h) - \text{UTCI}_{\text{baseline}}) \times \Delta h$$

where the summation is performed over the defined nighttime period. This produces a duration-weighted thermal load rather than relying on a single nighttime temperature value.

Decision 7 — Local Nighttime Baseline

The nighttime baseline will be derived from the historical climatology of the selected location rather than using an arbitrary universal temperature such as 28°C or 30°C. For the prototype, a specific percentile-based baseline method will be selected after inspecting the available historical ERA5 data.

The exact baseline period and percentile will be documented once the actual historical dataset is finalized. The resulting baseline value will then be frozen for reproducible

prototype analysis. In future development, the baseline methodology can be calibrated against local heat-health outcomes.

Decision 8 — Cumulative Thermal Burden: CTBI

We propose a **Cumulative Thermal Burden Index (CTBI)** to represent accumulated heat exposure over consecutive days. The purpose of CTBI is to capture the difference between an isolated hot day and a sequence of stressful days with persistent nighttime thermal load.

The proposed formulation is:

$$\text{CTBI}(t) = \alpha \times \text{CTBI}(t-1) + \beta \times \text{DayStress}(t) + \gamma \times \text{NCTL}(t)$$

Before combination, DayStress and NCTL will be normalized to comparable scales. The parameters α , β and γ are treated as tunable parameters rather than scientifically fixed constants. During a future health-data calibration stage, these parameters can be optimized using historical health outcomes.

For the historical analysis, CTBI will be calculated separately for each continuous year or heatwave period so that unrelated calendar periods are never treated as consecutive days.

Decision 9 — Vulnerability as Model Features

Vulnerability information will not be multiplied directly with CTBI using arbitrary weights. Instead, vulnerability indicators will be treated as features for the health-impact/risk layer.

Potential features include elderly population proportion, outdoor-worker exposure, population density, informal-settlement indicators, healthcare accessibility, vegetation and built-up characteristics. The actual features will depend on the availability and spatial resolution of reliable data.

Decision 10 — Response Capacity

Response capacity will be kept conceptually separate from heat-health risk. A ward can have high heat risk but relatively strong response capacity, while another ward can have similar heat risk but limited healthcare access or cooling infrastructure.

Therefore, the prototype will present response-capacity indicators separately and use them to support prioritization and action recommendations rather than hiding them inside an arbitrary weighted score.

Decision 11 — Health-Impact/Mortality Risk

The problem statement explicitly requires an automated mortality-risk component. However, a scientifically meaningful supervised mortality or hospitalization model requires appropriate historical health outcome labels.

Therefore, the prototype will include a transparent **rule-based Health-Impact/Mortality Risk Proxy** to demonstrate the intended decision layer without fabricating epidemiological probabilities. This proxy will be explicitly labelled as non-calibrated and non-validated.

Once ward-level or sufficiently granular hospitalization/mortality data becomes available, the same feature pipeline can be used to train and calibrate a supervised health-impact model.

Decision 12 — Machine Learning

Supervised ML will be used only when appropriate health outcome labels are available. Candidate models include XGBoost, Random Forest and Logistic Regression as a baseline comparison.

The intended input features include CTBI, NCTL, UTCI, humidity, consecutive hot days and spatial vulnerability indicators. The intended output is the probability of a high health impact over the next 3–5 days.

Synthetic labels will not be used to claim model accuracy, AUROC, F1 or mortality prediction performance.

Decision 13 — Explainability

For a future trained ML model, SHAP will be used to explain individual ward-level predictions and identify the main contributing features. SHAP visualizations will only be labelled as model-derived when they actually originate from a trained model. Any illustrative explanation shown before model training will be clearly marked as illustrative.

Decision 14 — Spatial Resolution

ERA5 and other meteorological datasets may have a coarser spatial resolution than individual municipal wards. Therefore, the system will include an explicit spatial-processing stage in which weather information is mapped or interpolated onto the selected ward polygons.

The final ward-level feature should therefore represent the best available spatial estimate rather than incorrectly claiming that the original ERA5 grid itself is ward-resolution data.

Decision 15 — Dashboard Technology

A lightweight web dashboard such as Streamlit will be used for the prototype because the objective of the current stage is to demonstrate the complete decision workflow rather than build production-scale infrastructure. The dashboard will eventually display ward-level heat risk, thermal burden, vulnerability, response capacity, explanation and recommended actions.

Decision 16 — Prototype Maturity Labels

Throughout the project, outputs will be classified into three maturity levels. **Demonstrated** means the component has been implemented using real or verified data. **Prototype** means the component is implemented as a demonstrative framework but requires further calibration or validation. **Future** means the component requires additional institutional data, operational access or long-term validation.

This distinction is intentional and will be maintained throughout the presentation so that the project does not overclaim capabilities that cannot yet be scientifically validated.

DOCUMENT 2 — PROBLEM & EXISTING SYSTEM RESEARCH NOTE

File name: 02_Problem_Existing_Systems_Research

PS 26083 — Problem Understanding & Existing-System Review

Problem Context

Extreme heat is not determined only by maximum air temperature. Human thermal stress can be affected by the combined influence of temperature, humidity, wind and radiation. A temperature threshold alone therefore provides limited information about how stressful the conditions may be for different populations and locations.

The problem statement specifically highlights the need to move from forecasting what the weather will be to understanding what that weather may do to human health. Our proposed system follows this principle by transforming meteorological information into human thermal-stress indicators and then combining the resulting thermal burden with population vulnerability and response capacity.

Current Warning Paradigm

Conventional heat-warning workflows generally begin with meteorological observations or forecasts and classify heat conditions using predefined criteria. Such warnings are important for public safety, but the information available to authorities can remain relatively general when the objective is to decide which specific wards, population groups or facilities require priority intervention.

Our project therefore does not aim to replace official meteorological warnings. Instead, it proposes an additional health-impact decision layer that can consume meteorological warnings and translate them into localized risk and response information.

Gap Identified

The key gap addressed by our project is the integration of several dimensions that are often handled separately. These include human thermal stress, multi-day cumulative heat exposure, persistent nighttime thermal load, spatial vulnerability, response capacity and actionable intervention.

The proposed contribution is therefore not a claim that UTCI, GIS, machine learning or heat-health warnings individually represent new inventions. The contribution is the

integration of these components into a single localized decision-support pipeline, with nighttime cumulative thermal load as the project's signature feature.

Nighttime Heat Motivation

Daytime peak heat does not fully describe prolonged heat exposure. When nighttime thermal conditions remain elevated, the population may continue to experience substantial thermal load after the daytime peak has ended. Our proposed NCTL metric operationalizes this idea by measuring the duration and magnitude of nighttime UTCI above a locally derived baseline.

NCTL should be interpreted as a proposed indicator of persistent nighttime thermal load and reduced opportunity for cooling, rather than as a clinically validated measurement of physiological recovery. Establishing its quantitative relationship with hospitalization or mortality would be part of the future validation stage.

Why Ahmedabad

Ahmedabad is being used as the pilot context because it provides a relevant environment for demonstrating heat-health decision support and connecting the proposed system with an established heat-action planning context. The pilot is intended to show how a city-level heat signal can be transformed into localized ward-level prioritization and intervention recommendations.

The city should not be presented as the only possible deployment location. The architecture is designed to be transferable to other Indian cities after appropriate local calibration and data integration.

Differentiation

The proposed system is differentiated by its end-to-end decision flow:

Weather → Human Thermal Stress → Nighttime Thermal Load → Cumulative Thermal Burden → Vulnerability → Health-Impact Risk → Explanation → Response Priority → Action.

The system therefore focuses not only on detecting heat but on helping authorities answer a sequence of operational questions: how severe is the thermal stress, how persistent is it,

which locations are more vulnerable, what response capacity exists, why is the risk elevated, and what action should be taken?

DOCUMENT 3 — DATA AVAILABILITY & DATA STRATEGY

File name: 03_Data_Availability_Strategy

PS 26083 — Data Availability & Validation Strategy

Historical Weather Data

The primary historical weather source for the prototype is ERA5 reanalysis. The data will be used to obtain hourly meteorological conditions required for thermal-stress computation and historical heat-event analysis. The current technical pipeline is being developed around real ERA5 data rather than synthetic weather observations.

Forecast Weather Data

The final early-warning system requires an operational forecast source for 3–5 day predictions. The intended sources are appropriate operational products from IMD/NCMRWF. Access and technical integration will be verified separately from the historical ERA5 pipeline.

If operational forecast access is unavailable during the prototype submission period, the project will use historical replay to demonstrate that the same feature-engineering pipeline can operate on forecast-like future time windows. This will be explicitly labelled as a prototype limitation rather than presented as live forecasting.

Ward Boundaries

Ward-level analysis requires reliable municipal boundary polygons in GeoJSON or Shapefile format. Ahmedabad ward boundaries will therefore be verified before the spatial pipeline is finalized.

If exact ward boundaries are unavailable in a suitable public format, the project will fall back to the highest reliable spatial unit available, such as municipal zones or another documented administrative boundary. The fallback will be clearly labelled instead of presenting a coarse spatial layer as actual ward-level information.

Population and Vulnerability Data

Potential vulnerability variables include elderly population proportion, population density, outdoor-worker exposure, informal-settlement indicators, healthcare accessibility and environmental characteristics such as vegetation and built-up density.

The project will use only variables for which a reliable source and appropriate spatial resolution can be established. When a variable is unavailable at ward level, the limitation will be documented and the feature may be moved to a future integration stage rather than replaced with fabricated values.

Environmental Data

Vegetation and built-up characteristics may be incorporated using remotely sensed or geospatial datasets such as NDVI or built-up indicators. These variables are intended to provide environmental context for identifying areas that may have lower natural cooling or higher exposure.

Because the current submission is time constrained, environmental enrichment will be prioritized after the core ERA5 → UTCI → NCTL → CTBI pipeline is working correctly.

Health Outcome Data

Health outcome data represents the most important dependency for scientifically calibrated health-impact and mortality prediction. The ideal dataset would contain historical heat-related hospitalizations or mortality observations at a sufficiently fine temporal and spatial resolution.

If ward-level health data is unavailable, city- or district-level health data may provide a lower-resolution validation pathway. If no appropriate health outcome data is available during the submission stage, the project will demonstrate a transparent rule-based risk proxy while clearly identifying supervised health-impact calibration as future work.

Synthetic health labels will not be presented as real observations and will not be used to claim model performance.

Validation Strategy

The thermal-stress pipeline will first be validated computationally using real weather data and sanity checks. NCTL and CTBI behaviour will be tested across continuous heat periods and compared with expected temporal patterns.

The eventual health-impact model will require backtesting against real historical health outcomes. Appropriate evaluation metrics may include AUROC, F1-score, recall, precision and calibration measures depending on the final target definition and class distribution.

The current prototype will distinguish between computational validation of the weather/thermal pipeline and epidemiological validation of health outcomes.

Problem Context and Mortality Burden

Extreme heat is a leading cause of weather-related mortality in India, with recent studies estimating that a single day of extreme heat causes approximately 3,400 excess deaths nationally, while a five-day heatwave can result in nearly 30,000 excess deaths. District-level analysis indicates that Ahmedabad alone experiences an estimated 307 excess deaths during a typical five-day heatwave, highlighting the severe localized impact. However, official heatstroke counts often significantly underreport the true burden, as many heat-related deaths are recorded as cardiovascular or respiratory failures. This discrepancy underscores the urgent need for a forecasting system that moves beyond simple temperature monitoring to predict actual physiological stress and mortality risk.[frontiersin+3](#)

Existing Solutions: Ahmedabad Heat Action Plan and IMD Warnings

Currently, disaster management relies on systems like the India Meteorological Department (IMD) district-level heatwave warnings and municipal Heat Action Plans (HAPs). The Ahmedabad Heat Action Plan, established after the devastating 2010 heatwave, is a pioneering example that combines early warnings with public health interventions, credited with reducing heat-related mortality by approximately 23%. However, these existing systems primarily trigger alerts based on dry-bulb temperature thresholds. While effective for general awareness, they often fail to account for the compounding effects of humidity, wind speed, and solar radiation on the human body. Furthermore, they typically operate at a district level, lacking the granularity to identify specific high-risk wards within a city.[confluence.ecmwf](https://confluence.ecmwf.org/en/heatwaves)

The Critical Gap: Nighttime Heat and Cumulative Exposure

A critical limitation of current early warning systems is their focus on daytime peak temperatures while ignoring the growing danger of nighttime heat. Research indicates that nighttime temperatures in India are rising faster than daytime temperatures, with over 70% of districts experiencing five or more additional very warm nights per summer season. Unlike daytime heat, persistent nighttime heat prevents the human body from recovering physiologically, leading to cumulative thermal stress that significantly increases mortality risk from cardiovascular and kidney strain. Studies show that mortality risk from nighttime heatwaves has increased significantly in recent years, yet most Heat Action Plans do not systematically incorporate nighttime thermal load or cumulative multi-day exposure into their alert triggers.

Scientific Basis: UTCI and Human Thermal Stress

To address these gaps, our system utilizes the Universal Thermal Climate Index (UTCI) rather than ambient temperature alone. UTCI is a comprehensive metric that integrates air temperature, humidity, wind speed, and mean radiant temperature (radiation) to estimate actual human thermal stress. This approach aligns with global health guidance, which emphasizes that combined heat-humidity metrics are superior predictors of physiological strain compared to temperature alone. By calculating UTCI, the system can distinguish

between a dry 40°C day and a humid 40°C day, accurately reflecting the higher mortality risk associated with high humidity and low wind conditions that impair the body's ability to cool itself.[impriiindia](https://www.impriiindia.org/)

Proposed Solution: Cumulative Burden and Ward-Level Prioritization

Our proposed solution introduces a novel operational layer that calculates a Cumulative Thermal Burden Index (CTBI) by integrating daytime thermal stress with a Nighttime Cumulative Thermal Load (NCTL) metric. This allows the system to quantify the "heat debt" accumulated by the population over multiple days, rather than treating each day in isolation. Unlike existing district-level alerts, our platform downscales this data to the ward level, enriching it with demographic vulnerability factors (such as elderly population density) and response capacity indicators (such as healthcare access). This enables the system to generate a dynamic "Heat-Health Priority" score, guiding authorities to deploy targeted interventions—such as opening cooling centers or shifting work hours—specifically to the wards most likely to experience severe health impacts.